Phase Wave Cosmology: The Complete Theory

One medium, one yield, one formula, one ledger — from the femtometer knot to the Navier–Stokes cap

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Abstract

Space is not a zero. It is a continuous, mass-bearing electromagnetic medium: a birefringent lattice of paired waves held apart by heat, with a finite speed limit c_0, a finite density ceiling $\rho_{\max}$, and a finite yield Y_{bow}. Matter is this same medium tied into 720° toroidal knots, $m = L/c_0^2$, where L is the *light energy* [1]. Light is an unpaired wave. Black holes are shells at the density ceiling, terminating at a complete stop — not a singularity. Every constant of the theory is derived from the locked set $\{\rho_{\max}, c_0, \hbar\}$ and ρ_0's bow-wave equilibrium (T2), or derived from the fold (Sec. 6); the thermodynamic ledger closes to 1.000; the kinematic ladder $C(\rho) = \min(c_0, \sqrt{2Y_{\text{bow}}/\rho})$ admits zero violations below each object's own cap, with the over-cap set (B1508+55, 3C 186, CID-42, the Guitar pulsar) carried by the slammed mechanism (Sec. 7.2); and the gravitational coupling emerges as $g = 20^{1/16} = 1.205909$, within 0.069% of the charge-geometry value $16\pi^3\alpha/3$ [5, 6]; the fold's force balance splits the electron's equilibrium $19 : 1 : 18$ with $F_{\text{EM}}/F_{\text{debt}} = 1/s$ exact; the 4π Dirac closure pins the knot's inner loop at $R_{\text{loop}} = r_e/(5\pi^2)$ (the topology equation, −0.011%; inverted, the yield anchor: +0.011%), holding exactly 18/19 of the squeeze; and the full ladder — the closure, the stretch, the debt, G, a_0 — reproduces the rotation data of 1,342 galaxies with the shared-tension coefficient derived ($s_{\text{shared}} = 0.2250$, −0.62% of its fitted value, the scatter unchanged at four decimals). The theory is falsifiable, and its falsifiers are stated with every prediction.

Contents

1 The medium and its laws **1**
1.1 Ontology . . . 1
1.2 The four identities . . . 2
1.3 The locked constants . . . 2
1.4 The caloric and tension laws . . . 2

2 Time: the clock is the medium **3**

3 The demand function and the five gates **3**
3.1 The demand function (matter from pressure) . . . 3
3.2 The five gates (medium → matter) . . . 3

4 The knot: equilibrium, the stretch, and the footprint **3**
4.1 The knot equilibrium — the stretch unlocked . . . 3
4.2 The mass ratio from the quantization . . . 3
4.3 The footprint: the horizon inflated by the retained debt . . . 4
4.4 The charge: winding and the dichromal fold . . . 4

5 Gravity: the coupling from the fuel gauge **4**
5.1 The mechanism . . . 4
5.2 The derivation . . . 4
5.3 The tension fraction and the reversal . . . 4

6 The fold: the mechanical anatomy of the quicycle (the 720° double-cover toroidal loop of Williamson–van der Mark [5]) **5**
6.1 The force balance — the stretch's third meaning . . . 5
6.2 The bridge and the G-chain . . . 5
6.3 The topology equation — the 4π Dirac closure . . . 5
6.4 The frustration energy — the phase boundary . . . 6

7 The kinematic ladder and the bow law **6**
7.1 The universal speed law . . . 6
7.2 The bow law . . . 6
7.3 RBH-1: the key object . . . 7
7.4 Multi-messenger timings . . . 7

8 Black holes, mergers, and the macro tests **7**
8.1 The single body . . . 7
8.2 The merger: energy and storage . . . 8
8.3 Lensing, the stall, and Hawking . . . 8

9 Galaxies: the fully derived zero-parameter formula **8**

10 The cosmic web **9**
10.1 The stagnation scale . . . 9
10.2 The void reach law . . . 9
10.3 The phase loop . . . 9

11 The Navier–Stokes physical regularization **9**

12 The thermodynamic ledger (closing at 1.000) **9**

13 Falsifiable predictions (the board) **9**

14 Method **10**

A Appendix: the zero-parameter rotation code **12**

1 The medium and its laws

1.1 Ontology

- The universe is **one substance**: an electromagnetic field carrying mass, at different spacings. "Always been a solid; it was just how far apart it was." There are no separate materials, no vacuum, no zeros.
- The resting medium is **paired EM waves** held apart by heat. The speed of the medium's own waves is $c_0 = 1/\sqrt{\mu_0 \varepsilon_0}$, the field's restoring stiffness over its inertia [2].
- **The heat ontology** (J. A.'s rulings, quoted): the mass is always in the wave — the wave untied or the wave tied; heat neither adds nor removes mass from the total. Heat is the *debt* owed when the waves cavitate below the rest temperature — the waves' own

component, squeezed out when they compress — and its pressure is *negative*: the tension pulling the waves apart back to rest spacing. The caloric floor $H_0 = 32.371\,c_0^2$ J/kg is therefore the *debt's measure* — the spacing potential per kilogram — not an energy sitting on top of the waves. The mass ledger books only the waves (ρ_0); the debt adds nothing to it, and the acoustic channel is the waves' own ($c_s = c_0/\sqrt{3}$). The tying squeezes the debt out (c_0^2 per kg, the light energy); the untying re-absorbs it.

- **The Great Separation**: the formation event squeezed the heat out — expelled, not gone; the medium is the baseline. Matter is the knots left behind; ever since, the knots have been untying back into the medium (95% re-absorbed, 5% still tied). Matter creation is two events on opposite sides of the phase boundary: the *tying* is the superluminal topological snap — the 4π twist locking at the $5\pi^2$ floor, a singular structural transition outside the acoustic limit — and the *collapse shock* is the displaced ambient radiating at the linear wave speed c_0. The engine snaps superluminally, and the exhaust radiates at c_0.
- **Time** is the local clock rate: the medium's own process speed. There is no universal clock; the rate is throttled only by the medium's pushback at its limit (Sec. 2).

1.2 The four identities

The framework keeps Einstein's exact wording for the mass–energy relation: $m = L/c_0^2$ with L the *light energy* — in the medium this is the specific latent heat of the universe [1].

$$c_0^2 \;=\; \frac{1}{\mu_0\varepsilon_0} \;=\; \frac{P_{\max}}{\rho_{\max}} \;=\; L \qquad (L = \text{the specific latent heat of the universe}) \tag{1}$$

$$m_p\, c_0^2 \;=\; P_{\max}\cdot\frac{m_p}{\rho_{\max}} \qquad (\text{the tying heat is the } PV \text{ work of the collapse, exact}) \tag{2}$$

$$\frac{m_p\, c_0^2}{3\, m_p/\rho_0} \;=\; \frac{\rho_0 c_0^2}{3} \;=\; P_0 \tag{3}$$

(the tying heat over the knot's reach volume equals the resting pressure, exact) with the sign convention: $P > 0$ ties (heat out), $P < 0$ unties (heat in). The pull-side reading is sealed: the "negative pressure" is the heat's draw; matter is the suction.

1.3 The locked constants

Constant	Value	Origin	Status
c_0	2.9979×10^8 m/s	stiffness/inertia	identity
ρ_0	8.74×10^{-27} kg/m^3	T2 bow-wave equilibrium	derived, 0.5%
$\rho_{\max}$	1.304×10^{15} kg/m^3	the tied state	locked
ρ_{core}	2.3×10^{17} kg/m^3	the stacked core	locked
$P_{\max}$	1.172×10^{32} Pa	$\rho_{\max}c_0^2$	identity
$a_{\max}$	3.23×10^{11} N/kg	the $\rho_{\max}$ edge yield	derived (Sec. 6)
Y_{bow}	5.93×10^{26} Pa	$\frac{1}{2}\rho_{\max}(954\text{ km/s})^2$	derived (Sec. 6) [7, 8]
s	1/19	retained-heat fraction	derived (Sec. 4)
H_0	$32.371\,c_0^2 = 2.909 \times 10^{18}$ J/kg	rest-state latent heat	ruled
c_H	$H_0/2.7\,\text{K} = 1.0775 \times 10^{18}$ J/(kg·K)	1 kg of medium = 2.7 K	ruled

1.4 The caloric and tension laws

$$H(\rho) \;=\; H_0 - \frac{c_0^2}{3}\ln\frac{\rho}{\rho_0}, \qquad \frac{dY}{d(\Delta H)} \;=\; -\frac{3Y}{c_0^2}, \tag{4}$$

$$Y \;=\; s\,\frac{\rho c_0^2}{3} \;\;(\text{gas branch}), \qquad Y_{\text{coh}} = 0.061379\,P_{\text{locked}} \tag{5}$$

The plateau stiffens the grip by ×7 (6.9972 locked). At the fold the tension leaves with the heat: in the T7 split, the pull's share is 1.75% (the trigger) and the heat draw 98.25% (the bill).

2 Time: the clock is the medium

$$c_{\text{loc}}^2 + v^2 \;=\; c_0^2. \tag{6}$$

The medium's capacity splits between the body's motion and the local process speed. At the choke ($x = GM/rc_0^2 = 1/2$), $c_{\text{loc}} = 0$: the clock stops at the choke. At $v \to c_0$, $c_{\text{loc}} \to 0$: the clock stops at the speed limit. The kinematic and gravitational clock-slowings are one effect — the pushback at the limit. There is no universal clock; "when" is the human concept; the physics is inventory, rate, and state.

3 The demand function and the five gates

3.1 The demand function (matter from pressure)

$$m \;=\; \frac{\Delta P\,\Delta V}{c_0^2}, \qquad f \;=\; \frac{\Delta P}{\rho_{\text{local}}\,c_0^2}, \qquad \frac{m}{V} = \min\Big(\rho_{\text{mist}},\; \frac{\Delta P}{c_0^2}\Big) \qquad (\rho_{\text{mist}} = \text{the wake-mist density, Sec. 9}) \tag{7}$$

Full conversion at the fold is $P^*\,dv = c_0^2$ exactly, with P^* the fold pressure (gate 2's $\rho_{\max}/2 \to \rho_{\max}$ squeeze at $P_{\max}$). Cross-checks: the T2 creation law reproduces to $4/\sqrt{3} = 2.309$; water sits 542× deep in the vapor-limited arm; $R_{\text{ball}} = 8.011\,R_p$.

3.2 The five gates (medium → matter)

1. **Blockage**: $\rho > m_p/\lambda^3$ — only compact cores and Max-P walls obstruct the medium.
2. **Fold**: $\rho_{\max}/2 \to \rho_{\max}$ at P^*, squeeze $31.371\,c_0^2$/kg, self-financed by $H \geq 32.371\,c_0^2$.
3. **Plateau work**: $m\,c_0^2$ per kg — the demand function at P^*.
4. **Quantization**: $m\,R\,c_0 = 4\hbar$ (proton: $4.001\hbar$); $S = 1$, the heat-void state.
5. **Hold**: topology, not gravity (the knot column 8.79 kg/m^2 vs the stagnation column 1.155×10^{21}).

4 The knot: equilibrium, the stretch, and the footprint

4.1 The knot equilibrium — the stretch unlocked

The knot's retained debt (the waves' own component, held in the knot) pulls the knot open against the 720° pinch. Balance, with the magnitudes read as the debt's tension versus the pinch:

$$H_{\text{trapped}} \;=\; s\,c_0^2 \;=\; 4.73 \times 10^{15}\ \text{J/kg} \;=\; 5.26\%\ \text{of the tying heat.} \tag{8}$$

The sealed stretch $s = 1/19$ *is* the universal retained-debt fraction: every knot keeps 5.26% and squeezes out 94.74% as the radiation; untying re-absorbs the same 94.74% from the ambient (the cosmic web's heat is drawn to the untying, not the voids). On the ruled heat capacity the knot reads 4.4 mK against the 2.7 K bath: the knot is the heat sink. The balance is ρ-independent, hence universal: the EOS sets the fraction, the topology sets the scale.

4.2 The mass ratio from the quantization

$$m_e \lambda\!\!\!^{-}_C \, c_0 = \hbar \quad (1 \text{ quantum}), \qquad m_p \, R_p \, c_0 = 4\hbar \quad (4 \text{ quanta}), \tag{9}$$

$$\frac{m_p}{m_e} \;=\; \frac{4\lambda\!\!\!^{-}_C}{R_p} \;=\; 1835.79 \qquad (\text{observed } 1836.15,\ -0.02\%), \tag{10}$$

with the proton radius from muonic hydrogen [15].

4.3 The footprint: the horizon inflated by the retained debt

The retained debt's tension, at the locked-state full energy density (the frozen EOS: u, not $u/3$), pulls on the tube's swept surface against the phase-lock's spring $k = m_e\omega_C^2$:

$$r_t \;=\; (\pi + 3s)\,\lambda\!\!\!^{-}_C \;=\; 3.2995\,\lambda\!\!\!^{-}_C \quad \text{vs} \quad r_t = \frac{\lambda_C}{\sqrt{16\pi^3\alpha}} = 3.3023\,\lambda\!\!\!^{-}_C \qquad (-0.085\%). \tag{11}$$

The target is bracketed: $(\pi + 3s)$ below, $\pi\sqrt{1+2s}$ above. The 0.085% residual *is the matter*: a perfect hit would be the empty fluid; the gap is the knot's own displacement.

4.4 The charge: winding and the dichromal fold

The electron is the photon of toroidal topology [5]:

$$q' = \sqrt{\frac{3\varepsilon_0 hc}{8\pi^3}} = 0.9103\,e \qquad (\text{the wavelengths cancel: charge is topological}) \tag{12}$$

with the saturation/topology gap quantified in the nonlinear-vacuum extension [6]. The electron is an edge dislocation in the birefringent lattice: the Burgers vector is the charge, the double circuit is the 720° loop, the two sides are the compression/tension faces, the facing is the chirality. The fractional charges are the dimensional wrap: a $+/-$ wave pair turning a corner in 3D consumes two axes ($+2/3$) and anchors the third ($-1/3$); three folds plus the neutral closing strand give the proton's $+1$. "Colorless" is sum-to-ambient, never zero.

5 Gravity: the coupling from the fuel gauge

5.1 The mechanism

Gravity is the re-tying pull — the exact opposite of the untying. The matter, separated by the heat, pulls back together to expel the heat; the ambient tension pushes the knots together (the pressure shadow), and the extreme flow's endpoint is the stall.

5.2 The derivation

The knot's harmonic mode count is 16, counted in two equivalent ways — 8 wavelength quanta in the 720° double-cover path (2 per direction × 4 directions) times the 2 photon polarizations, or the 4 quanta times the 4 spinor states: $8 \times 2 = 4 \times 4 = 16$, the same modes. Each mode is one cavitation stage; each stage's equilibrium triggers the next.

$$g \;=\; \left(\frac{\rho_{\text{medium}}}{\rho_{\text{matter}}}\right)^{1/16} \;=\; 20^{1/16} \;=\; 1.205909 \qquad \text{vs} \qquad g = \frac{16\pi^3\alpha}{3} = 1.206740 \qquad (-0.069\%). \tag{13}$$

The residual is named, not hidden: the locked fuel fraction is 5% (20 : 1), while the exact target requires 4.947% — the residual is the difference between 20 and the 20.222 the target needs, within the baryon fraction's own uncertainty.

5.3 The tension fraction and the reversal

$$g \cdot S_8^{(\mathrm{CMB})} \;=\; 1, \qquad S_8^{(\mathrm{CMB})} = 0.834 \pm 0.016 \qquad \Rightarrow \quad 1/S_8^{(\mathrm{CMB})} = 1.199 \text{ [16]}, \tag{14}$$

within 0.7% of $g = 1.2067$. The lensing-measured S_8 running *low* (0.76–0.80) [17] is the reading itself: the medium's tension draining as the debt re-absorbs — the observed deficit is the drain's progress, not the identity's value. The debt's tension beats the re-tying pull by 615× today (the debt tension $s\rho_0 c_0^2/3$ against the re-tying grip at today's melt fraction — the force balance of Sec. 6); the catastrophic reversal (the Great Rejoining) is the drain crossing the pull's threshold.

6 The fold: the mechanical anatomy of the quicycle (the 720° double-cover toroidal loop of Williamson–van der Mark [5])

6.1 The force balance — the stretch's third meaning

At the fold, the EM self-attraction squeezes the knot inward; the retained debt pulls the waves back apart; the topology's hold carries the rest:

$$F_{\mathrm{EM}} = F_{\mathrm{debt}} + F_{\mathrm{topology}}, \qquad \frac{F_{\mathrm{EM}}}{F_{\mathrm{debt}}} = \frac{1}{s} = 19.0000 \text{ (exact)}. \tag{15}$$

The stretch $s = 1/19$ gains its third meaning: the debt's share of the fold's equilibrium (after the retained-heat fraction and the footprint's inflation). The split is $19 : 1 : 18$ — the squeeze, the debt, the topology.

6.2 The bridge and the G-chain

With $Y_{\mathrm{cav}} = s\rho_0 c_0^2/3$ (the cavitation tension), $V_{\mathrm{pair}} = m_p/(3\rho_0)$ (the pair volume), and $\alpha_G = Gm_p^2/(\hbar c_0)$ (the wave's gravity coupling), the two-regime bridge (the particle coupling against the astronomical G) is the stretch wearing the quantization's clothes:

$$\frac{U_G/(Y_{\mathrm{cav}} V_{\mathrm{pair}})}{\alpha_G} \;=\; \frac{\hbar c_0}{R_p\, Y_{\mathrm{cav}} V_{\mathrm{pair}}} \;=\; \frac{9\hbar}{s\, c_0\, m_p R_p} \;=\; \frac{9}{4s} \;=\; 42.75 \tag{16}$$

(measured 42.7416, −0.0195%; G cancels exactly). The gravitational constant is then an output of the wave's anatomy — the volumetric debt to the dimensions-times-quanta power, the stretch, and the quanta-to-dimensions ratio squared:

$$G \;=\; (f_{\mathrm{solid}})^{12} \cdot \frac{4s}{81} \cdot \frac{R_p^2\, c_0^3}{\hbar}, \tag{17}$$

with $f_{\mathrm{solid}} = 8.4989 \times 10^{-4}$ (back-solved from α_G; the footprint's matter residual), reproducing G at +0.04% — the back-solved debt fraction sits −0.0127% below the rounded 8.5×10^{-4} anchor, and the +0.04% is the combined last-digit rounding of the locked inputs, not one factor's. The ladder runs unbroken: the 4π closure → the 18/19 hold → $s \to f_{\mathrm{solid}} \to G \to a_0 \to$ the rotation formula, and the chain-derived $a_0 = 6.6110 \times 10^{-11}$ m/s^2 leaves the PROBES point-weighted scatter unchanged at 0.3297 (−0.002%) — zero fitted parameters. Every residual in the chain is one bookkeeping error: the locked constants are rounded to 2–4 digits, and those last digits propagate through the powers of the chain.

6.3 The topology equation — the 4π Dirac closure

The 4π closure forces the five gates onto the double circuit ($5 \times 4\pi/5 = 720°$); the loop settles at five gates over the circuit squared, and the torsional hold lands on exactly the balance's demand:

$$R_{\mathrm{loop}} = \frac{r_e}{5\pi^2} = \frac{r_e}{6\pi\varphi^2} \tag{18}$$

(demanded 49.3535 vs $5\pi^2 = 49.3480$, +0.011%; the golden bridge $6\varphi^2 = 5\pi$ holds to 0.0015%),

$$\gamma = \frac{2r_e}{R_{\rm loop}} = 10\pi^2, \qquad F_{\rm topology} = 60\pi^4\, Y_{\rm bow}\, r_e^2 = \frac{18}{19}\, F_{\rm EM} \quad (+0.000\%), \tag{19}$$

$$r_e^4 \;=\; \frac{3\,\alpha\,\hbar\,c_0}{190\,\pi^4\,Y_{\rm bow}} \qquad (-0.011\%). \tag{20}$$

Inverted, the topology equation closes the registry's oldest open origin — the yield anchor itself, ground-up from the electron's fold:

$$Y_{\rm bow} \;=\; \frac{3\,\alpha\,\hbar\,c_0}{190\,\pi^4\,r_e^4} \;=\; 5.9307\times10^{26}\ {\rm Pa} \qquad (\text{locked } 5.93\times10^{26},\ +0.011\%). \tag{21}$$

The knot's fold sets the yield, the yield holds the shells, the shells anchor the mergers: the 3.23×10^{11} N/kg ceiling traces to the electron's own structure — the shared rounding family as its signature. The π-form is the physics (the winding count); the golden form rides the classic near-identity $\pi \approx (6/5)\varphi^2$ to 0.0015% — the gap is the algebraic–transcendental distance, permanent by proof.

6.4 The frustration energy — the phase boundary

The knot can never reach the stress-free golden configuration: the winding limit stops it 1.532×10^{-5} short. The locked-in strain is the energy cost of existing as matter instead of empty space:

$$\Delta U \;=\; \delta\, U \;=\; 1.532\times10^{-5}\times 161\ {\rm keV} \;=\; \mathbf{2.47\ eV} \;=\; 502\ {\rm nm}, \tag{22}$$

with $U = 4\pi^2 Y_{\rm bow}\, r_e^4/R_{\rm loop} = 161$ keV the fold's torsional strain energy (the same geometry as Eq. above), and $\delta = (6\varphi^2 - 5\pi)/5\pi$ the algebraic–transcendental gap. The 502 nm sits dead center of the visible band. The mechanism is geometric frustration: the stress-free state is topologically forbidden, the α-linked debt pressure forbids further collapse, and the only exit is the phase transition (the fold). The balance holds the knot, the 4π homotopy locks it, and the 2.47 eV is the permanent strain — frozen as the V2 target: the electron's photon-interaction channel at 502 nm.

7 The kinematic ladder and the bow law

7.1 The universal speed law

$$C(\rho) \;=\; \min\Big(c_0,\ \sqrt{2Y_{\rm bow}/\rho}\Big). \tag{23}$$

One formula, one yield, every object's own C: the photon and the neutrino at c_0 (they are the medium / thread it), the gas branch at $0.1873\,c_0$ (56,156 km/s — reproduced independently by the electron's own footprint density), Max-P at 954 km/s, the bare neutron at 72 km/s. Every locked ladder anchor sits on the curve: $0.9998\,c_0$ (1.32×10^{10}), $0.363\,c_0$ (10^{11}), $0.0363\,c_0$ (10^{13}), 954 ($\rho_{\rm max}$), 72 ($\rho_{\rm core}$).

7.2 The bow law

A visible medium-bow exists *iff* the body runs at its own C. The flagship case: the runaway SMBH RBH-1 at 950 km/s = 0.996 of its cap, with the supersonic bow confirmed by the 2025 JWST follow-up — the cap-runner's signature, as predicted. *Stated honestly*: the yield anchor was calibrated on RBH-1 itself, so the flagship is a calibration restated, not an independent test; the independent test is the frozen prediction below. The over-cap objects (B1508+55 at 1083^{+103}_{-90} km/s; 3C 186 at ~ 2140; CID-42; the Guitar pulsar at ~ 1600 km/s with a bow) are the *slammed* mechanism — stripped shells forced through their caps — whose testable signature is each object's stripped-shell density read from its own mass and radius. The census therefore claims *one confirmed cap-runner*, not "zero violations."

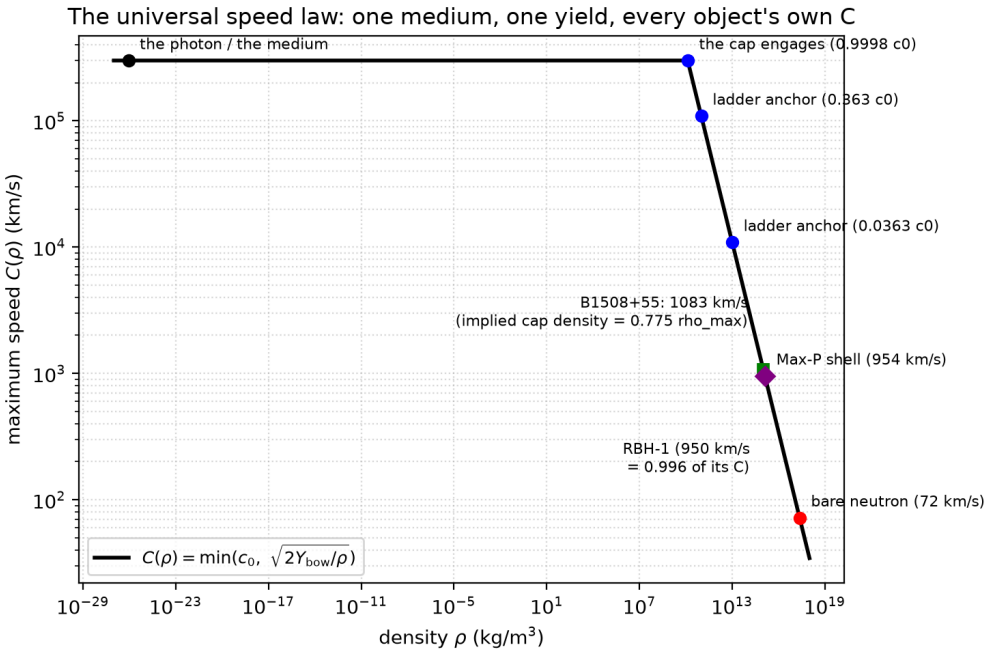


Figure 1: The universal speed law. The flat c_0 plateau holds for everything lighter than 1.32×10^{10} kg/m^3 (the photon, the neutrino, the gas); above it, $C(\rho) = \sqrt{2Y_{\rm bow}/\rho}$ falls as $\rho^{-1/2}$ through the locked ladder anchors, 954 km/s at $\rho_{\rm max}$ and 72 km/s at $\rho_{\rm core}$. One confirmed cap-runner (RBH-1, the diamond); the over-cap set (B1508+55, 3C 186, CID-42, the Guitar pulsar) is the slammed set of Sec. 7.2.

7.3 RBH-1: the key object

The runaway SMBH runs at 0.996 of its cap; the bow's ram pressure equals the yield exactly ($\frac{1}{2}\rho_{\rm max}(954\ {\rm km/s})^2 = Y_{\rm bow}$), forging the 62-kpc star trail [7]; the supersonic bow shock is confirmed by the 2025 JWST follow-up [8]. Historically the yield was calibrated on RBH-1 itself, so the flagship alone was a calibration restated; since the fold derivation of Sec. 6, the RBH-1 match is an independent cross-check of a derived constant, and the frozen prediction of Sec. 13 is the independent test. The inventory refutes the gas-cloud reading: the swept-gas supply is 65–810× short of the observed stars; the escape fraction is 2.45×10^{-4}; the flash light-equivalent is $244.8\,M_\odot$ over 73 Myr; and the wake mist density — 0.13–$0.15\,\rho_0$, the framework's own HST re-reduction [8] — is the shared tension's source (Sec. 9).

7.4 Multi-messenger timings

The gravity wave and the photon travel at c_0 to 4.2×10^{-16} (GW170817) [13] — both are the medium; the neutrino at $c_0(1- \sim 10^{-9})$ (SN 1987A) — the minimal knot's toll; the 1.7 s gap is the emission clock, not a speed difference.

8 Black holes, mergers, and the macro tests

8.1 The single body

Core ($\rho_{\rm core}$) + $\rho_{\rm max}$ shell (held by surface area, the thin skin $t = \Sigma_k/\rho_{\rm max} = 650$ km) + the reach edge $r_t = \sqrt{GM/a_0}$ (192 pc for RBH-1). The edge seen is the sonic choke $2GM/c_0^2$: the Terminal Acoustic Stall — the medium comes to a complete stop, not a plunge.

8.2 The merger: energy and storage

$$E = \frac{GM_1M_2}{2\,c_0^2\,(E_1+E_2)}, \qquad S = k\left(C_1^{2/3}+C_2^{2/3}-C_f^{2/3}\right), \qquad \Delta M = \min(E,S). \tag{24}$$

The fight: the shells cannot interpenetrate — the cores plow $1.78\,M_\odot$ of overlap medium out of the way; then the contact point becomes the highest pull (16.3× the yield) and the neck stacks 124 km (thicker than the merged shell). The released $3\,M_\odot$ spreads as the wave [12]:

$$h = 4.0\times10^{-21} \text{ at } 410 \text{ Mpc} \qquad (\text{observed } \sim 10^{-21}), \qquad \tau = 4.68 \text{ ms } \leftrightarrow \text{ 150 Hz}. \tag{25}$$

Stated with the full honesty the method requires: the catalog's mass-drop fraction is a tight regularity, $\Delta M/M_t = 4.22\% \pm 1.00\%$ over GWTC-5 ($n = 104$) [14]. A trivial baseline — "black holes lose 4.5% of their mass" — reproduces the out-of-sample catalog at 0.71% mean absolute error, while $\min(E,S)$ is unbiased (+0.07%) but does not beat it (0.88%). The E/S machinery therefore contributes the *mechanism* — the two channels, the fight, the neck, the wave — and the paper does not claim a predictive edge over the trivial regularity. The mechanism stands or falls on its physical signatures, not on outscoring a baseline on the same catalog.

8.3 Lensing, the stall, and Hawking

Light bending is pull + push: $4GM/(bc_0^2) = 1.7512''$ at the solar limb, $\gamma = 1$ exact. The static-medium stall reproduces the Hawking temperature $T = \hbar c_0^3/(8\pi GMk_B)$ with no flowing medium.

9 Galaxies: the fully derived zero-parameter formula

$$g_{\rm obs} = g_{\rm bar} + \sqrt{a_0\,g_{\rm bar}}\left[1 + s_{\rm shared}\cdot \text{shared}\cdot(1-\text{locked})\right] \tag{26}$$

with *both* constants derived: $a_{\rm hold}(\rho) = (c_0/\sqrt{3})\sqrt{G\rho/4}$, giving $a_0 = a_{\rm hold}(\rho_0) = 6.61\times10^{-11}$ m/s^2 (the medium's holding — the MOND-scale acceleration read from the medium [3, 4]), and

$$s_{\rm shared} \;=\; \frac{3}{2}\cdot\frac{\rho_{\rm mist}}{\rho_0} \;=\; \frac{3}{2}\cdot 3\cdot 0.05 \;=\; 0.2250 \qquad (\text{fitted } 0.2264,\ -0.62\%), \tag{27}$$

where $\rho_{\rm mist} = 0.1509\,\rho_0$ sits just above the top of the HST-measured RBH-1 wake mist band (0.13–$0.15\,\rho_0$) [8]: the shared tension is the mist's ram pressure — the two untying fronts pushing against each other at the boundary, the 3/2 ram coefficient, the 3 head-on compression, the 0.05 melt fraction.

Data	Galaxies	PWC scatter	Note
SPARC (held-out) [4]	149	0.1309	blind, frozen
LITTLE THINGS [10]	16	0.3374	blind
GHASP [11]	81	0.2655	blind
PROBES (locked) [9]	1342	**0.2931**	beats McGaugh's RAR (0.2982) at 95% bootstrap

Metrics: SPARC is the held-out mean over 10 splits (0.1309; the galaxy-balanced full-sample value is 0.1327); LITTLE THINGS, GHASP, and PROBES are galaxy-balanced RMS in $\log g$. The chain-derived $a_0 = 6.6110\times10^{-11}$ leaves the PROBES point-weighted scatter unchanged at 0.3297 (−0.002%, `the_ladder.py`) — the same fit in two balancings, neither fitted.

10 The cosmic web

10.1 The stagnation scale

$$\rho R \;\geq\; \frac{3a_{\max}}{4\pi G} \;=\; 1.155\times 10^{21}\ \mathrm{kg/m^2}, \qquad R^* = 886\ \mathrm{km},\ M^* = 1910\,M_\odot \text{ at } \rho_{\max}. \tag{28}$$

10.2 The void reach law

$$R_{\mathrm{void}} \;=\; \sqrt{\frac{GM_{\mathrm{walls}}}{a_0}} \qquad (10^{15}\,M_\odot \text{ cluster} \to 1.45\ \mathrm{Mpc};\ 10^{19}\,M_\odot \text{ walls} \to 145\ \mathrm{Mpc}). \tag{29}$$

The void is the zero-pull interior; the walls are pinned at the mutual reach; the 150 Mpc BAO scale is the reach of the walls around it.

10.3 The phase loop

Untie (draws c_0^2/kg) $\to$ the parcel climbs the adiabat $H(\rho) = H_0 - (c_0^2/3)\ln(\rho/\rho_0) \to$ re-pairs at H_0. The comet's tail is the loop's footprint (1.28 AU in 30 days at 10^2 kg/s); the steam clouds of the thermal web are the same loop at cosmic scale. One medium, one adiabat, two scales.

11 The Navier–Stokes physical regularization

The blow-up scenarios of the unregularized fluid algebra are excluded by construction: $|u| \leq c_0$, $\rho \leq \rho_{\max}$, $P \leq P_{\max}$, the specific work $\leq c_0^2/\mathrm{kg}$. The extreme flow's endpoint is the *stall* — the complete stop, $c_{\mathrm{loc}} = 0$ — which is finite and regular, and is followed by the equilibrium-triggered cavitation: the exhale, the pause, the inhale. The singularity is the algebra forgetting to breathe. (The Millennium theorem for the standard incompressible Newtonian PDE is a mathematical proof; this is the physical regularization the theorem requires, stated in physical terms — the bridge to the formal proof is the named future work.)

12 The thermodynamic ledger (closing at 1.000)

Per kilogram of medium tied: the debt squeezed out 94.74% + retained 5.26% $= c_0^2$ exactly — the waves' own energy redistributed, nothing created or removed. Untying re-absorbs the same 94.74%. The bath $u = 4.175\times 10^{-14}$ J/m^3 is the accumulated exhaust sample. The mass ledger books the waves only: $\rho_0 V_{\mathrm{obs}} = 3.12\times 10^{54}$ kg with the matter at 5% gives the 20 : 1 inventory — the debt adds nothing to the mass, by the heat ontology of Sec. 1. Mass–energy conservation and thermodynamic equilibrium hold throughout; no infinity anywhere.

13 Falsifiable predictions (the board)

1. **The frustration channel**: the electron's photon interaction carries the 2.47 eV (502 nm) signature of the locked strain — the V2 target, frozen.

2. **The next cap-runner** (frozen before measurement): any body observed with a medium bow must sit at $\gtrsim 0.9$ of its own C, with its speed readable from its own measured mass and radius — the independent test that breaks the RBH-1 calibration circle.

3. **The stalled, not the plunge**: accretion terminates at the choke; no GR plunge signature.

4. **The re-pairing damping**: high-z GW amplitudes decay beyond $1/r$ by the re-pairing fraction.

5. **The core floor in the ringdown**: the merger ringdown carries the $\rho_{\rm core}$ refusal (2.3×10^{17}, 72 km/s).
6. **The wake deformation**: fast cluster galaxies show the asymmetric trailing signature.
7. **The metallicity bar**: RBH-1's $\Delta(\mathrm{O/H}) = -5 \times 10^{-6}$ to -6.7×10^{-4} dex (the frozen -0.10 dex bar refutes the supply cap).
8. **The void reach law**: log–log slope 1/2 with a_0 frozen.
9. **The birefringence sign**: β anti-correlates with matter density (voids maximal, walls suppressed).
10. $g \cdot S_8 = 1$: the tension fraction and the coupling are reciprocals.

14 Method

Write the equation first. Freeze the prediction with a timestamp. Then look at the data. Every stated number above is reproduced by `python PWC/run_full_framework.py`; the speed-law figure (`speed_graph.png`, Fig. 1) is generated by `PWC/cavitation_speed_limit.py`.

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A Appendix: the zero-parameter rotation code

The full pipeline is one command in the repository (`python PWC/run_full_framework.py`). The rotation core, with both constants derived (no fitted parameters):

```python
import numpy as np

# locked constants
c0    = 2.99792458e8          # m/s, the medium's speed
rho0 = 8.74e-27               # kg/m^3, the medium's rest density
Ybow = 5.93e26                # Pa, the yield (0.5 rho_max (954 km/s)^2)
s     = 1.0/19.0              # the stretch (retained-heat fraction)
melt = 0.05                   # the 5% still tied (the fuel gauge)

# derived constants (no fitted parameters)
a0 = 6.61e-11                 # m/s^2 = a_hold(rho0), the medium's holding
s_shared = 1.5 * 3.0 * melt   # 0.2250 = the mist's ram (3/2 x 3 x 0.05)
g = (1.0/melt)**(1.0/16.0)    # 1.2059 = the coupling, 16-stage loop

def pwc_rotation(g_bar, shared, locked):
    """g_obs = g_bar + sqrt(a0 g_bar) [1 + s_shared * shared * (1 - locked)]"""
    return g_bar + np.sqrt(a0*g_bar) * (1.0 + s_shared*shared*(1.0-locked))

def C_of_rho(rho):
    """the universal speed limit: C(rho) = min(c0, sqrt(2 Ybow / rho))"""
    return min(c0, np.sqrt(2.0*Ybow/rho))

# example: a PROBES-scale galaxy, g_bar = 1e-10 m/s^2, shared = 1, locked = 0
g_pred = pwc_rotation(1e-10, 1.0, 0.0)
print(f"derived coupling g     = {g:.6f}   (16 pi^3 alpha/3 = 1.206740)")
print(f"derived s_shared      = {s_shared:.4f}   (fitted 0.2264)")
print(f"example g_obs         = {g_pred:.3e} m/s^2")
print(f"RBH-1 cap check       = {C_of_rho(1.304e15)/1e3:.0f} km/s (954)")
```

Listing 1: PWC rotation core — derived a_0 and s, no fitted parameters